

Finding The Forgotten Exhibition of Black Inventors and Artisans in 1851

By Michiko Quinones | Published: April 21, 2024 | Updated: August 13, 2025

Article Text

173 years ago—today (I decided to publish today—April 21—which was the last day of the fair), a Black organization called "The Coloured American Institute In the State of Pennsylvania for the Promotion of Mechanic Arts and Sciences" held an exhibition to highlight the creativity and innovation of Black artisans, designers and inventors. It was called "The Colored Mechanics Fair."

This was both a technical/scientific exhibition and an arts exhibition. The word "mechanic" today means someone who works on vehicles. But in 1851 the meaning of the word was different and included anyone who built objects with their hands.

While we don't have the names and entries of everyone who participated in the fair, which was judged, we do have the names of the winners thanks to a write up in the Friends Journal.

The Variety of Arts, Crafts and Technical Innovation is ☆☆☆Astonishing☆☆☆

There was technology: David Spraw displayed his "hydrometer for draining swamps". Dr. John Rock featured his handmade artificial teeth. Aaron Roberts provided a scaled down model of his machine to move railroad cars back onto their tracks.

There was art: David Bustill Bowser, John Venning and Alfred Wilnon had exhibits of oil and water color paintings. Jane Vasham showcased her wax fruit sculptures.

There was Design, Photography and Printing: Joseph Gardner displayed his maps and surveys. J.W. Johnson and Robert Douglass showed of their daguerreotypes. And Isaac Jackson bought his stereotype molds for printing.

Some of the participants have played important historical roles. Dr. John Rock was a well-known Dentist and lawyer. Dr. James J.G. Bias was a community leader and underground railroad agent. F.A. Duterte built coffins but his wife Henrietta learned his trade and eventually ran his shop, using coffins to hide freedom seekers on their journey north. And David Bustill Bowser is one of the most famous artists to emerge from the 1838 Black Metropolis.

There was so much that we think it's a lot to read so we created an online exhibit of the some of the fair participants for you here.

<https://1838-black-metropolis.shorthandstories.com/the-colored-mechanics-fair/index.html>

We encourage you to read all the winners in the Friends Journal. Note that this was just the winners, so there were lots more entrants than were listed here.

The Details

Black Technical Organizations

The Coloured American Institute is a bit of an enigma. We haven't been able to find more about it. We think it may have been an outgrowth of the Humane Mechanics' Society, which had 65 members in 1838.

The Humane Mechanics' Society was founded in 1827 and we have gleaned their purpose from a manuscript of a July 4 speech to the organization in the Leon Gardiner collection at the Historical Society of Pennsylvania, Box 2G.

It reads: "Arts are divided into two classes, useful or mechanic, liberal or polite. The former are known as trades such as black smiths, carpenter, brick layer, baker, weaver"

This speech is not signed and it seems slightly different from the July 4 Joseph Coor speech printed in Dorothy Porter's Early Negro Writings. However, we assume that since the audience and the date are the same, that this is a manuscript of Joseph Coor's speech.

So we can see from this definition that a wide swath of occupations would have been considered mechanical; blacksmiths, whitesmiths, tanners, metal workers etc.

Below please find an interactive visual of the industrial breakdown of trades derived from the 1838 Census.

Note that many people may have had various trades or artistic pursuits but they listed their primary job in the census. For example, as you see below, inventor Aaron Roberts, a participant and winner in this fair, listed himself as a "porter" in 1847.

But many Black people stepped proudly into a variety of trades and occupations in 1838, as you'll see from exploring this visual.

The Humane Mechanics' Society and the Colored American Institute both indicate that the Black community was organizing around trades and supporting technical creativity and innovation.

The story of Aaron Roberts makes that clear.

Aaron Roberts, Inventor

Aaron Roberts first appears as an inventor in the Coloured Mechanics Fair in 1851. At this time he was working on railroad inventions.

There is an Aaron Roberts in the 1838 census. He and his wife are not from Pennsylvania, but their child is. They both attended Little Wesley church and their child went to school. They lived in Ten Foot Alley, which was close to the 6th and Pine intersection.

Aaron Roberts later appears in the 1847 PAS Census and the 1850 US Census. In 1847 he is living at 6 Currant Alley and is a Porter. He is married but there are now two children.

The 1850 US Census lists his wife Rachel and a 20 year old daughter Rebecca. He is still listed as a Porter.

By 1855 he had moved onto developing a machine to extinguish fires. And he started to make national news. Here's a Kentucky newspaper excerpt:

The Franklin Institute took note. Their Committee on Science and the Arts said "The whole arrangement displays ingenuity" and it would "supersede the necessity of carrying the hose up on the ladder."

Note that at this time The Franklin Institute was still not allowing Black people in the building. Institute for Colored Youth Professor Robert Campbell was refused entry to The Franklin Institute in November 1856 due to "color". He wrote a letter to the institute:

"On last Monday evening I again applied, when after some attempt to evade a direct refusal I was told that no ticket could be sold to me. I inquired what the objections were, and received the very emphatic reply of 'color.' My object, gentlemen, in this note is simply to ascertain whether you endorse the action of your agent in his refusal."

While the white scientific community, as signified by The Committee of Science and the Arts at the Franklin Institute, recognized Roberts' ingenuity, it appears that they did not support him.

But the Black community did.

In 1858, the Massachusetts Colored Convention resolved to support him.

In 1859, the Banneker Institute supported him by creating a committee specifically to raise funds for him, essentially a 19th century go fund me, which held a fundraising concert to help him continue.

Roberts was recognized for his contributions to Fire Extinguishing history in a 1907 Encyclopedia. "The Water Tower" may be a colloquial name for his invention.

After this we haven't been able to find more mentions of Aaron Roberts' invention activities in the records.

The Fair and The Philadelphia Institute

There was a call for entries to the fair in the April 5, 1851 Public Ledger.

It was held at the Philadelphia Institute, 717 Lombard Street. We have previously made the case for a Black municipal center based on findings of two townhomes on 7th between Lombard and South, one of which is Benezet Hall. But this letter from Stephen Smith, after the burning of Beneficial Hall, indicates that there were three townhomes used for thousands of people in hundreds of societies that might have as many as 10 meetings per night.

The Philadelphia Institute was also where the Euphemia Williams celebration was held.

We think that 717 Lombard might be the third townhome Smith mentioned. It was around the corner from the other two townhomes and was used by a variety of groups from 1850 to the late 1870s. It was also across the street from the first site of the Institute for Colored Youth (720 Lombard) which raises the interesting possibility of a shared support for education and sciences with activities occurring in the two buildings. Note that the Institute for Colored Youth officially opened on Lombard the next year, 1852.

It was the headquarters of the Pennsylvania Equal Rights league and may have also been the publishing office of "The Palm and Pine" newsletter.

And the building was used to generate wealth in the Black community. The pattern appears to be that there are joint owners who issue stock which pays dividends based on rental of the property until the entire property is paid off. See this blog on Benezet hall for evidence of this pattern. This letter proves that the debt on the building was completely paid off.

Note that on this letter, Philadelphia Institute was three buildings—715, 716 and 717. What that means—since 716 was across

the street—is that the use of the term "institute" may have identified a whole complex of buildings; the way we call a whole complex of buildings a "campus".

This photo of Lombard Street above 7th is from 1903. 717 and 715 Lombard are no longer standing but you can see them clearly here.

The fair closed on April 21, 1851. And it got good press. Here's the April 10, 1851 write up in the Public Ledger:

Closing Thoughts

Even after two decades of increasing disenfranchisement, substandard public schools and mob attacks, the Black Metropolis still thrived, still maintained a political, social, and educational center at 7th and Lombard and still produced technical and scientific leadership.

Racism is a disease and a delusion that continues to drag on the technical innovation and creative possibilities of this country, today and in 1851. Thank God The Black Metropolis existed. Through events like this fair, Black creatives had support and uplift for the beauty of new ideas.

Sources

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Images

- A circa 1860 Wax Fruit Display made in Philadelphia. Similar to what Jane Vasham displayed at the Fair. Courtesy Philadelphia Museum of Art —
https://static.wixstatic.com/media/af048d_6163878478ef41a59c54eabc0dbc6ee9~mv2.jpg
- AI image of Aaron Roberts standing next a model of his inventions at the 1851 Coloured Mechanics Fair —
https://static.wixstatic.com/media/af048d_30120ed803f84b279e464134632e1fa3~mv2.jpg
- Excerpt from 'The Present State and Condition of the People of Color' —
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- Manuscript of July 4 Speech to the Humane Mechanics Society, courtesy Historical Society of Pennsylvania —
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- Aaron Roberts' entry in the Coloured Mechanics Fair, from the Friends Journal Article —
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- Excerpt from The Courier-Journal Louisville, Kentucky · Friday, September 21, 1855 —
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- Excerpt from Journal of the Franklin Institute —
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- Notes of a resolution from the 1858 Colored Convention supporting Aaron Roberts —
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- Banneker Institute Support for Aaron Roberts, from Tony Martin's article on the Banneker Institute —
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- Add for submissions to the fair, April 5, 1851 Public Ledger. Courtesy Newspapers.com —
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- Excerpt from the March 30, 1843 Anti-Slavery Standard, Courtesy Newspapers.com —
https://static.wixstatic.com/media/af048d_701ec0d63d764462bc4287339af6e5be~mv2.png
- Map showing the locations of the Halls near 7th and Lombard —
<https://www.arcgis.com/apps/instant/sidebar/index.html?appid=4dd29a66cd1344539bdfcca8219ef344>
- Pennsylvania Equal Rights League Minutes, Leon Gardiner Collection, Box 2G —

https://static.wixstatic.com/media/af048d_c401a15f36eb4dfc884223b51f564995~mv2.png

- Debt Paid notice from the Philadelphia Institute, Leon Gardiner Collection, Box 2G —
https://static.wixstatic.com/media/af048d_3260e12131664c1e8d64150fda3e2e03~mv2.png
- 1903 Photo of Lombard Street with map reference from 1860 Hexamer and Locher map —
https://static.wixstatic.com/media/af048d_0a7a51972c374130bd4b0a84737d712c~mv2.png

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